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Method for measuring amyloid β peptide

Abstract

Disclosed is a method for measuring an A β peptide, comprising: mixing a blood sample comprising an A β peptide and an antibody that specifically binds to the A β peptide to form a complex of the A β peptide and the antibody; releasing the AP peptide from the complex with a basic solution comprising an organic solvent; and measuring the released A β peptide by mass spectrometry.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority from prior Japanese Patent Application No. 2020-109825, filed on Jun. 25, 2020, entitled “Method for measuring A β peptide and reagent composition used in the method”, the entire contents of which are incorporated herein by reference.

FIELD OF THE INVENTION

(2) The present invention relates to a method for measuring amyloid β (A β) peptide.

BACKGROUND

(3) An A β peptide in a biological sample collected from a subject is known to be a biomarker for Alzheimer's disease. Since cerebrospinal fluid (CSF) contains a relatively large amount of A β peptide among biological samples, a method for quantitatively measuring an A β peptide in CSF has been established. For example, Leinenbach A. et al., Mass Spectrometry-Based Candidate Reference Measurement Procedure for Quantification of Amyloid- β in Cerebrospinal Fluid. Clinical Chemistry 60: 7 987-994 (2014) describes that an A β peptide in CSF is measured by liquid chromatography-mass spectrometry (LC-MS).

(4) Since collection of CSF is invasive, burden on a subject is large. Therefore, recently, a method for measuring an A β peptide using blood, which has a low collection burden, as a biological sample has been developed. However, since the amount of A β peptide contained in blood is less than that in CSF, a method for accurately measuring an A β peptide has been required. An object of the present invention is to provide a means for enabling accurate measurement of A β peptide in blood.

SUMMARY OF THE INVENTION

(5) The scope of the present invention is defined solely by the appended claims, and is not affected to any degree by the statements within this summary.

(6) The present invention provides a method for measuring an A β peptide, comprising mixing i) a blood sample containing an A β peptide and ii) an antibody that specifically binds to the A β peptide, to form a complex of the A β peptide and the antibody, releasing the A β peptide from the complex

with a basic solution containing an organic solvent, and measuring the released A β peptide by mass spectrometry.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram showing an example of a container containing a reagent composition;
- (2) FIG. 2 is a diagram showing results of measurement of A β peptide using LC-MS/MS;
- (3) FIG. 3A is a diagram showing concentrations of A β 40 peptide in plasma of each specimen;
- (4) FIG. 3B is a diagram showing concentrations of A β 42 peptide in plasma of each specimen;
- (5) FIG. 4A is a diagram showing a calibration curve prepared based on measured values detected using an A β 40 peptide with a known concentration;
- (6) FIG. 4B is a diagram showing a calibration curve prepared based on measured values detected using an A β 42 peptide with a known concentration;
- (7) FIG. 5 is a diagram showing difference in elution efficiency of A β peptide due to a difference in composition of free reagent;
- (8) FIG. 6A is a diagram showing difference in elution efficiency of A β 40 peptide due to a difference in organic solvent of free reagent;
- (9) FIG. 6B is a diagram showing difference in elution efficiency of A β 42 peptide due to a difference in organic solvent of free reagent;
- (10) FIG. 7 is a diagram showing a difference in amount of carryover due to a difference in composition of free reagent;
- (11) FIG. 8A is a diagram showing an amount of A β 40 peptide when a 190 pg/ml A β 40 peptide solution is eluted with a basic solution;
- (12) FIG. 8B is a diagram showing an amount of A β 40 peptide when a 190 pg/ml A β 40 peptide solution is eluted with an acidic solution;
- (13) FIG. 8C is a diagram showing an amount of A β 42 peptide when a 103 pg/ml A β 42 peptide solution is eluted with a basic solution;
- (14) FIG. 8D is a diagram showing an amount of A β 42 peptide when a 103 pg/ml A β 42 peptide solution is eluted with an acidic solution;
- (15) FIG. 9A is a diagram showing an amount of A β 40 peptide when a 50 pg/ml A β 40 peptide solution is eluted with a basic solution;
- (16) FIG. 9B is a diagram showing an amount of A β 40 peptide when a 50 pg/ml A β 40 peptide solution is eluted with an acidic solution;
- (17) FIG. 9C is a diagram showing an amount of A β 42 peptide when a 26 pg/ml A β 42 peptide solution is eluted with a basic solution; and
- (18) FIG. 9D is a diagram showing an amount of A β 42 peptide when a 26 pg/ml A β 42 peptide solution is eluted with an acidic solution.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (19) The method for measuring an A β peptide of the present embodiment is a method for measuring an A β peptide captured by an antibody by mass spectrometry (hereinafter, also referred to as "measurement method"). This measurement method is characterized in that the A β peptide is released from the complex of the A β peptide and the antibody with a basic solution containing an organic solvent, and the released A β peptide is measured by mass spectrometry.
- (20) Conventionally, when a test substance is released from the test substance captured by the antibody (that is, the complex of the antibody and the test substance), an acidic solution is used. This is because an acidic solution generally has an excellent action of dissociating a binding between a test substance and an antibody. For example, Japanese Laid-Open Patent Publication No. 2018-194374 describes that an A β peptide is released from the complex by an acidic aqueous

solution with a pH of 1 to 4, which may contain an organic solvent. However, as shown in the examples described later, the present inventors found that an acidic solution containing a released A β peptide is not suitable for mass spectrometry in combination with liquid chromatography. Specifically, when the acidic solution containing the released A β peptide was used as it was for liquid chromatography, a phenomenon in which a part of the charged A β peptides remained in a flow path and column of a liquid chromatography apparatus and the remaining A β peptide was carried over to measurement of next sample (called carryover) occurred. When carryover occurs, accurate measurement of A β peptide cannot be performed. In order to reduce carryover, it is conceivable to release the A β peptide in an acidic solution and then exchange the solvent with a basic solution. However, even if the carryover can be reduced by this method, it may be difficult to measure with high sensitivity because the A β peptide is lost due to solvent exchange. On the other hand, in the measurement method of the present embodiment using a basic solution containing an organic solvent, the amount of A β peptide carried over is significantly reduced. Therefore, even if the basic solution containing the released A β peptide is used as it is for liquid chromatography, the A β peptide can be accurately measured and no solvent exchange is required. As described above, since a blood sample has a lower content of A β peptide than CSF, the measurement method of the present embodiment is suitable for quantitative measurement of A β peptide in the blood sample.

(21) In the measurement method of the present embodiment, first, an antibody that specifically binds to an A β peptide is mixed with a blood sample containing an A β peptide to form a complex of the A β peptide and the antibody. A complex of an A β peptide and an antibody can be formed by mixing a blood sample containing an A β peptide with an antibody that specifically binds to the A β peptide.

(22) As used herein, the term “blood sample containing an A β peptide” also includes blood samples suspected of containing an A β peptide. Examples of the blood samples include blood (whole blood), plasma, serum, and the like. Of these, plasma and serum are preferred. The blood sample may be diluted with an appropriate aqueous medium as necessary. The aqueous medium is not particularly limited as long as it does not interfere with the measurement described later. Examples of the aqueous medium include water, physiological saline, a buffer solution, and the like. The buffer solution is a buffer solution having a buffering effect at a pH near neutrality (for example, a pH of 6 or more and 8 or less). Examples of the buffer solution include Good buffers such as HEPES, MES, and PIPES, tris buffered saline (TBS), phosphate buffered saline (PBS), and the like.

(23) An origin of the blood sample is not particularly limited. For example, the blood sample may be blood collected from a subject and plasma or serum prepared from the blood. Commercially available pool plasma, healthy person plasma or the like may be used. A labeled A β peptide as an internal standard substance may be added to the blood sample as necessary. The subject is not particularly limited. Examples of the subject include a healthy person, a person having abnormality in cognitive function, and a person suspected of having the abnormality. Examples of cognitive dysfunction include mild cognitive impairment (MCI), Alzheimer's disease, and the like.

(24) A β peptide is a polypeptide produced by treating amyloid β precursor protein (APP) with β -secretase and γ -secretase. Unless otherwise specified, A β peptides include polypeptides of any length, but are usually polypeptides consisting of 39 to 43 amino acids. As the A β peptide, A β 40

(25) (DAEFRHDSGYEVHHQKLVFFAEDVGSNKGAIIGLMVGGVV: SEQ ID NO: 1) consisting of 40 amino acid residues and A β 42

(26) (DAEFRHDSGYEVHHQKLVFFAEDVGSNKGAIIGLMVGGVVIA: SEQ ID NO: 2) consisting of 42 amino acid residues are preferable.

(27) The A β peptide may be in the form of a monomer or a multimer. A multimer, also called a polymer, is formed by physically or chemically polymerizing or aggregating a plurality of monomeric A β peptides. The multimer may contain a plurality of monomeric A β peptides, and may contain other molecules. In the multimer, the monomeric A β peptides do not need to be tightly

bound to each other by covalent bonds or the like. Multimers also include aggregates in which a plurality of monomeric A β peptides are aggregated by looser binding. Examples of the multimer of A β peptide include A β oligomers and the like.

(28) As used herein, the term “antibody” includes full-length antibodies and fragments thereof. Examples of the antibody fragments include Fab, Fab', F(ab')₂, Fd, Fd', Fv, light chain, heavy chain variable region (VHH) of heavy chain antibody, reduced IgG (rIgG), one chain antibodies (scFv), and the like. The antibody that specifically binds to an A β peptide may be either a monoclonal antibody or a polyclonal antibody, but is preferably a monoclonal antibody.

(29) The monoclonal antibody itself that can specifically bind to the A β peptide is known. Examples thereof include an antibody of clone 82E1 (called 82E1 antibody) that recognizes 1st to 16th regions counting from an N-terminal amino acid residue of the A β peptide as an epitope, an antibody of clone 6E10 (called 6E10 antibody) that recognizes 3rd to 8th regions as an epitope, an antibody of clone WO-2 (called WO-2 antibody) that recognizes 4th to 10th regions as an epitope, an antibody of clone 2H4 (called 2H4 antibody) that recognizes 1st to 8th regions as an epitope, an antibody of clone H31L21 (called H31L21 antibody) that recognizes 36th to 42nd regions as an epitope, an antibody of clone G2-11 (called G2-11 antibody) that recognizes 33rd to 42nd regions as an epitope, an antibody of clone 16C11 (called 16C11 antibody) that recognizes 33rd to 42nd regions as an epitope, an antibody of clone 21F12 (called 21F12 antibody) that recognizes 34th to 42nd regions as an epitope, an antibody of clone 1A10 (called 1A10 antibody) that recognizes 35th to 40th regions as an epitope, and the like. These monoclonal antibodies are commercially available.

(30) In the present embodiment, it is preferable to use an antibody that binds to one or both of A β 40 and A β 42 as an antibody that specifically binds to the A β peptide. Examples of the antibody that specifically binds to A β 40 include an antibody of clone 1A10. Examples of the antibody that specifically binds to A β 42 include antibodies of clones H31L21, G2-11, 16C11 and 21F12. Examples of the antibody that binds to both A β 40 and A β 42 include antibodies of clones 82E1, 6E10, WO-2 and 2H4. By using these antibodies, A β 40 and/or A β 42 can be selectively captured among the A β peptides in the blood sample. In this case, at least one of A β 40 or A β 42 can be released in releasing of the measurement method of the present embodiment, and a measured value of the at least one of A β 40 or A β 42 can be acquired in measuring.

(31) In a preferred embodiment, an antibody that binds to both A β 40 and A β 42 is used. In this case, A β 40 and A β 42 can be released in the releasing of the measurement method of the present embodiment, and a measured value of A β 40 and a measured value of A β 42 can be acquired in the measuring.

(32) In the measurement method of the present embodiment, it is preferable to form a complex of the A β peptide and the antibody on the solid phase in order to selectively acquire the A β peptide captured by the antibody. The complex can be formed on the solid phase by contacting a solution containing the complex with a solid phase on which the antibody can be immobilized. Alternatively, an antibody that specifically binds to the A β peptide may be previously immobilized on the solid phase and used. By using an antibody immobilized on the solid phase, the complex can be formed on the solid phase. Specifically, the complex is formed on the solid phase by mixing the solid phase on which the antibody that specifically binds to the A β peptide is immobilized and the blood sample. Then, the complex can be selectively acquired by separating an unreacted free component and the solid phase and recovering the solid phase.

(33) The solid phase may be any insoluble carrier capable of immobilizing an antibody. The mode of immobilization of the antibody on the solid phase is not particularly limited. For example, the antibody and the solid phase may be bound directly, or the antibody and the solid phase may be indirectly bound via another substance. Examples of the direct binding include physical adsorption and the like. Examples of the indirect binding include immobilizing a molecule that specifically binds to an antibody on a solid phase, and immobilizing the antibody on the solid phase through

binding between the molecule and the antibody. Examples of the molecule that specifically binds to the antibody include protein A or G, an antibody (a secondary antibody) that specifically recognizes an antibody, and the like. A combination of substances interposed between the antibody and the solid phase can be used to immobilize the antibody on the solid phase. Examples of the combination of substances include combinations of any of biotin and its analogs and any of biotin-binding sites, a hapten and an anti-hapten antibody and the like. The biotin and its analogs include biotin and biotin analogs such as desthiobiotin and oxybiotin. The biotin-binding sites include avidin and avidin analogs such as streptavidin and tamavidin (registered trademark). Examples of the combination of a hapten and an anti-hapten antibody include a combination of a compound having a 2,4-dinitrophenyl (DNP) group and an anti-DNP antibody. For example, by using an antibody previously modified with biotin or its analog (or a compound having a DNP group) and a solid phase to which a biotin-binding site (or anti-DNP antibody) is previously bound, the antibody can be immobilized on the solid phase through binding between the biotin or its analog and the biotin-binding site (or binding between the DNP group and the anti-DNP antibody).

(34) The material of the solid phase is not particularly limited. For example, the material can be selected from organic polymer compounds, inorganic compounds, biopolymers, and the like. Examples of the organic polymer compounds include latex, polystyrene, polypropylene, styrene-methacrylic acid copolymer, styrene-glycidyl (meth)acrylate copolymer, styrene-styrene sulfonate copolymer, methacrylic acid polymer, acrylic acid polymer, acrylonitrile butadiene styrene copolymer, vinyl chloride-acrylate copolymer, polyvinyl acetate acrylate, and the like. Examples of the inorganic compounds include magnetic bodies (iron oxide, chromium oxide, cobalt, ferrite, etc.), silica, alumina, glass, and the like. Examples of the biopolymer include insoluble agarose, insoluble dextran, gelatin, cellulose, and the like. Two or more of these may be used in combination. The shape of the solid phase is not particularly limited. Examples of the shape of the solid phase include a particle, a microplate, a microtube, a test tube, and the like. Among them, particles are preferable, and magnetic particles are particularly preferable. When a particle is used as the solid phase, a complex forming in the measurement method of the present embodiment corresponds to a general immunoprecipitation method.

(35) Subsequently, in the measurement method of the present embodiment, the A β peptide is released from the complex with a basic solution containing an organic solvent (hereinafter, also referred to as "free reagent"). The free reagent is considered to have an action of dissociating the binding between the antibody and the A β peptide. In a preferred embodiment, the solution containing the complex formed on the solid phase is mixed with the free reagent. As a result, the A β peptide is released from the complex, and the released A β peptide and the solid phase on which the antibody is immobilized are present in the mixed solution. For example, when the solid phase is a magnetic particle, a solution containing the A β peptide can be recovered by separating the solid phase on which the antibody is immobilized and the solution containing the A β peptide by centrifugation or magnetic separation.

(36) Conditions for contact between the complex and the free reagent are not particularly limited. For example, the free reagent at a temperature of 4° C. or more and 42° C. or less is contacted with the complex and allowed to stand or agitated for 1 minute or more and 10 minutes or less. An amount of the free reagent used is not particularly limited. For example, the amount can be appropriately determined, for example, from the range of 10 μ L or more and 50 μ L or less per sample.

(37) The free reagent can be obtained by mixing a basic solution with an organic solvent, or by mixing water with a basic substance and an organic solvent. The basic solution can be obtained by mixing water and a basic substance. A commercially available basic solution such as aqueous ammonia may be used. Examples of the basic substance include a substance that donates an ammonium ion and the like. Examples of the substance that donates an ammonium ion include ammonia, ammonium carbonate, ammonium bicarbonate, and the like. The basic substance in the

free reagent may be one type or two or more types.

(38) Examples of the organic solvent include acetonitrile, acetone, 1-propanol, 2-propanol, hexane, ethanol, dimethyl sulfoxide, methanol, and the like. The organic solvent in the free reagent may be one type or two or more types. In the present embodiment, the free reagent preferably contains at least acetonitrile as the organic solvent, and more preferably contains only acetonitrile as the organic solvent.

(39) The concentration of the organic solvent in an elution reagent is not particularly limited, and is preferably 20% or more, 30% or more, or 40% or more. The concentration of the organic solvent is preferably 65% or less, 60% or less, or 55% or less. The concentration “%” of the organic solvent as used herein is all volume/volume % (v/v %).

(40) A pH of the free reagent is not particularly limited as long as it is a pH recognized as basic by those skilled in the art. The pH of the free reagent is preferably 10.85 or more, 10.9 or more, 10.95 or more, 11.0 or more, 11.05 or more, 11.1 or more, 11.15 or more, 11.2 or more, 11.25 or more, 11.3 or more, 11.35 or more, or 11.4 or more. Most preferably, the free reagent has a pH of 11.4 or more. As a result, the A β peptide can be released from the complex particularly efficiently. The pH of the free reagent is preferably 14.0 or less, 13.5 or less, 13.0 or less, 12.5 or less, 12.4 or less, 12.35 or less, 12.3 or less, 12.25 or less, 12.2 or less, 12.15 or less, 12.1 or less, 12.05 or less, or 12.0 or less.

(41) The pH of the free reagent can be adjusted by an amount (or concentration) of basic substance added. When ammonia or a salt thereof is used as the basic substance, the molar concentration of ammonium ions in the free reagent is preferably 5.29 mM or more, 10.57 mM or more, 26.43 mM or more, 52.85 mM or more, 105.71 mM or more, or 132.14 mM or more. The molar concentration is preferably 1586 mM or less, 1533 mM or less, or 1480 mM or less. For example, when the basic substance is ammonia, the concentration of ammonia in the free reagent is preferably 0.01% or more, 0.02% or more, 0.05% or more, 0.1% or more, 0.2% or more, or 0.25% or more. The concentration is preferably 3% or less, 2.9% or less, or 2.8% or less. The concentration “%” of ammonia as used herein is all weight/weight % (w/w %).

(42) The measurement method of the present embodiment may include washing step of removing an unreacted free component that has not formed a complex between the complex forming and the releasing. This washing includes B/F (Bound/Free) separation. The unreacted free component is a component that does not constitute a complex, and examples thereof include antibodies that have not bound to an A β peptide. The washing method is not particularly limited, and in a case where the complex is formed on a solid phase, when the solid phase is a particle, the complex and the unreacted free component can be separated by recovering the particle by centrifugation or magnetic separation, and removing supernatant. When the solid phase is a container such as a microplate or a microtube, the complex and the unreacted free component can be separated by removing a liquid containing the unreacted free component. After removing the unreacted free component, the solid phase capturing the complex may be washed with a suitable aqueous medium such as PBS.

(43) Then, in the measurement method of the present embodiment, the released A β peptide is measured by mass spectrometry. The mass spectrometry is not particularly limited as long as the released A β peptide can be measured, and a known ionization capable of measuring the A β peptide can be used. Examples of the ionization include electrospray ionization (ESI), atmospheric chemical ionization (APCI), matrix-assisted laser desorption ionization (MALDI), and the like. Among them, the ESI method is particularly preferred.

(44) The mass spectrometer used in the mass spectrometry is not particularly limited, and the mass spectrometer can be appropriately selected from known mass spectrometers. Examples thereof include a quadrupole (Q) mass spectrometer, ion trap (IT) mass spectrometer, a flight time (TOF) mass spectrometer, a Fourier transform ion cyclotron resonance (FTICR) mass spectrometer, an IT-TOF type mass spectrometer, a Q-TOF type mass spectrometer, a triple quadrupole (QqQ) type mass spectrometer, and the like. Among them, it is preferable to measure using a mass spectrometer

capable of MS/MS measurement, and more preferably to measure using a triple quadrupole mass spectrometer.

(45) In mass spectrometry, liquid chromatography-mass spectrometry (LC-MS), which is a combination of a mass spectrometer and a liquid chromatography apparatus, is preferably used, and LC-MS/MS which is a combination of liquid chromatography with a mass spectrometer capable of MS/MS measurement is preferably used.

(46) The liquid chromatography apparatus is not particularly limited as long as it can be connected to a mass spectrometer, and a commercially available high performance liquid chromatography (HPLC) apparatus can be used. The column connected to the liquid chromatography apparatus is not particularly limited, and a commercially available column for HPLC can be used. In the present embodiment, the A β peptide released in the free reagent can be subjected to a liquid chromatography apparatus. Therefore, pre-measurement process such as solvent exchange is not required.

(47) The column is not particularly limited, but for example, a reversed-phase column can be used. Examples of a filler of the reversed-phase column include a silica-based filler, a polymer-based filler, and the like. Among them, a silica-based filler is preferred. As the reversed-phase column having a silica-based filler, a basic-resistant ODS column is more preferred.

(48) As the solvent (mobile phase) used for liquid chromatography, a solution having the same composition as the free reagent or a basic solution used for the free reagent can be used. The concentration of organic solvent in the mobile phase is not particularly limited, and the concentration can be appropriately set according to measurement conditions. Liquid chromatography may be performed using an isocratic method using a mobile phase with a single concentration. Alternatively, liquid chromatography may be performed using a stepwise method or a gradient method in which a plurality of mobile phases with different compositions are used in combination.

(49) Flow velocity of the mobile phase in the measurement can be appropriately set according to properties such as materials of the apparatus and the column and pressure resistance. The flow velocity of the mobile phase is preferably set to a flow velocity at which mass spectrometry is appropriately performed. The column temperature, the amount of a sample introduced into a measuring apparatus and the like can be appropriately set according to the measuring apparatus and the column.

(50) In the measurement method of the present embodiment, measurement may be performed by a liquid chromatography apparatus to acquire information about the sample. Examples of the detector that performs measurement include a UV detector, a fluorescence detector, a differential refractive index detector, an electrical conductivity detector, an electrochemical detector, and the like.

(51) The measurement of A β peptide by mass spectrometry can be appropriately set from a known technique according to the ionization and the mass spectrometer to be used. Among them, when a triple quadrupole mass spectrometer is used, it is preferable that the A β peptide is measured by multiple reaction monitoring (MRM) measurement in which the measurement mode is set to positive ion measurement mode.

(52) The triple quadrupole mass spectrometer has a first quadrupole (Q1) and a third quadrupole (Q3) in front of and behind a collision cell (Q2). An object to be measured is ionized by an ion source to be precursor ions, and only ions having a specific mass-to-charge ratio (m/z) set in Q1 pass through a filter and are introduced into Q2. The precursor ions with a specific mass-to-charge ratio introduced into Q2 collide with an inert gas and cleave to be product ions. The product ions sent from Q2 to Q3 are filtered again in Q3, only ions with a specific mass pass through the filter and are sent from Q3 to a detector, and a signal is detected. As a result, a specific product ion for a specific precursor ion is detected by the detector. A combination of the specific mass-to-charge ratio set in Q1 and the specific mass-to-charge ratio set in Q3 is called an MRM transition. In MRM measurement, by setting multiple MRM transitions, multiple product ions can be detected at

the same time. As a result, a plurality of substances contained in the object to be measured can be measured at the same time. For example, each MRM transition is set for a sample containing A β 40 and A β 42, whereby two types of A β peptides can be measured at the same time and each measured value can be acquired.

(53) In the MRM measurement, the measurement mode is not particularly limited, but it is preferable to use the mass spectrometer in the positive ion measurement mode. Cone voltage and collision energy in the MRM measurement are also not particularly limited. The cone voltage and collision energy can be set to appropriate conditions by those skilled in the art.

(54) For example, as an MRM transition for A β 40, precursor ion/product ion can be set to 1083/1953.6. As an MRM transition for A β 42, precursor ion/product ion can be set to 1129/1078.5. By setting the MRM transitions as described above, the A β peptide can be detected with high sensitivity.

(55) In the measurement method of the present embodiment, the blood sample may be mixed with a labeled internal standard substance, and the measured value of the A β peptide may be acquired based on the measured value of the internal standard substance. The internal standard substance is not particularly limited as long as it can quantify the A β peptide, but the internal standard substance is preferably an A β peptide labeled with a stable isotope. Examples of the A β peptide labeled with a stable isotope include A β 40 and A β 42 labeled with N15 stable isotope. The amount of the internal standard substance added is not particularly limited as long as the concentration can be measured by the measuring apparatus. The amount of the internal standard substance added can be appropriately set according to analytical ability of the measuring apparatus.

(56) In addition to the MRM transition for detecting the A β peptide, an MRM transition can be set and measured for the A β peptide labeled with a stable isotope as the internal standard substance, and the measured value can be acquired. As the MRM transition for A β 40 labeled with N15 stable isotope, precursor ion/product ion can be set to 1096/1066.5. As the MRM transition for A β 42 labeled with N15 stable isotope, precursor ion/product ion can be set to 1142.5/1091.5.

(57) A further embodiment is a reagent composition used in the measurement method of the present embodiment. This reagent composition is a basic solution containing an organic solvent. The basic solution containing an organic solvent is as described above. By using the reagent composition, the A β peptide can be released with high efficiency from a complex of the A β peptide and the antibody that specifically binds to the A β peptide.

(58) A pH of the reagent composition is not particularly limited as long as it is a pH recognized as basic by those skilled in the art. The pH of the reagent composition is preferably 10.85 or more, 10.9 or more, 10.95 or more, 11.0 or more, 11.05 or more, 11.1 or more, 11.15 or more, 11.2 or more, 11.25 or more, 11.3 or more, 11.35 or more, or 11.4 or more. Most preferably, the reagent composition has a pH of 11.4 or more. As a result, the A β peptide can be released from the complex particularly efficiently. The pH of the reagent composition is preferably 14.0 or less, 13.5 or less, 13.0 or less, 12.5 or less, 12.4 or less, 12.35 or less, 12.3 or less, 12.25 or less, 12.2 or less, 12.15 or less, 12.1 or less, 12.05 or less, or 12.0 or less.

(59) The reagent composition may contain additives such as stabilizers to an extent that it does not affect the release of A β peptide from the complex and the measurement of A β peptide by mass spectrometry. Examples of the additive include bovine serum albumin (BSA), human albumin, egg white albumin, monosaccharides such as glucose, disaccharides such as maltose, sugar alcohols such as mannitol and sorbitol, amino acids such as glycine, and the like. The additive may be one type or two or more types. The additive is preferably contained in the range of 0 parts by weight or more and 20 parts by weight or less based on 100 parts by weight of the basic solution containing an organic solvent. The reagent composition may be contained in a suitable container that does not affect composition of the reagent composition. FIG. 1 is a diagram showing an example of a container **151** containing the reagent composition.

(60) Hereinafter, the present disclosure will be described more specifically with reference to

Examples.

EXAMPLES

Example 1: Measurement of Plasma A β Peptide Using Combination of Immunoprecipitation Using Basic Solution Containing an Organic Solvent and Mass Spectrometry

(61) An A β peptide was released from a complex of an A β peptide and an antibody that specifically binds to the A β peptide using a basic solution containing an organic solvent, and the A β peptide was measured using mass spectrometry.

(1) Capture and Release of A β Peptide Using Immunoprecipitation

(1.1) Blood Sample

(62) As blood samples containing an A β peptide, 5 types of commercially available plasma samples (ProMedeX) from different lots were used.

(1.2) Antibody that Specifically Binds to A β Peptide

(63) As an antibody that specifically binds to the A β peptide, 6E10 antibody (BioLegend, Inc.) which is a commercially available mouse monoclonal anti-A β antibody was used. The 6E10 antibody was immobilized on a magnetic particle (M-270 Epoxy-activated Dynabeads: Thermo Fisher Scientific Inc.) by a conventional method.

(1.3) A β Peptide

(64) An A β 40 peptide and an A β 42 peptide were purchased from AnaSpec, Inc. for preparation of a calibration curve. As internal standard substances, 15N-A β 40 and 15N-A β 42 (rPeptide) which were an A β 40 peptide and an A β 42 peptide each labeled with a stable isotope ^{15}N were used. The A β 40 peptide was suspended in PBS solutions containing 3% BSA, so as to have final concentrations of 10.8 pg/ml, 21.7 pg/ml, 43.3 pg/ml, 86.6 pg/ml, 173.2 pg/ml, 346.4 pg/ml and 692.8 pg/ml, respectively. The A β 42 peptide was suspended in PBS solutions containing 3% BSA, so as to have final concentrations of 2.8 pg/ml, 5.6 pg/ml, 11.3 pg/ml, 22.6 pg/ml, 45.2 pg/ml, 90.3 pg/ml and 180.6 pg/mL, respectively. 15N-A β 40 and 15N-A β 42 were suspended in the same solution in PBS solutions containing 3% BSA so as to be 500 pg/ml, respectively.

(1.4) Preparation of Basic Solution Containing Organic Solvent

(65) As a basic solution (free reagent) containing an organic solvent, 1.2 ml of 28% concentrated ammonia water (Nacalai Tesque, Inc.) and 6.0 ml of acetonitrile (Kanto Chemical Co., Inc.) were added and mixed to 12.8 ml of pure water to prepare a 1.68% ammonia/30% acetonitrile solution.

(1.5) Immunoprecipitation

(66) A 250 μl of plasma sample or each of the A β 40 peptide solutions or A β 42 peptide solution prepared in (1.3) above was added to a 1.5 ml sample tube (Eppendorf AG). To each sample tube containing the above solution was added 250 μl of the solution containing 15N-A β 40 and 15N-A β 42 prepared in (1.3) above, and the sample tube was allowed to stand at room temperature for 30 minutes. After standing the sample tube, 40 μl of the suspension of magnetic particles (4 μg antibody/0.4 mg magnetic particles) immobilized with 6E10 antibody prepared in (1.2) above was added to each sample solution, and the mixture was inverted and mixed for 1 hour using a rotator at room temperature to form a complex of the A β peptide and the antibody. These solutions were focused using a magnetic stand to remove supernatant.

(1.6) Washing

(67) After removing the supernatant, 1 mL of a PBS solution containing 3% BSA was added to the magnetic particles remaining in the sample tube, mixed, and then magnetized again to remove supernatant. This operation was performed twice with 1 mL of the PBS solution containing 3% BSA, twice with 1 mL of a 50 mM ammonium acetate solution and once with 1 mL of ultrapure water successively to wash the magnetic particles.

(1.7) Release of A β Peptide from Complex

(68) After washing the magnetic particles in (1.6) above, 25 μL of the free reagent prepared in (1.4) above was added to the remaining magnetic particles after removing the washing liquid, mixed, and the mixture was allowed to stand for 1 minute. The magnetic particles were magnetically collected

again, and supernatant was recovered as an eluate.

(2) Mass Spectrometry

(69) The eluate prepared in (1.7) above was subjected to LC-MS/MS for MRM measurement, and the A β peptide was measured. ACQUITY (registered trademark) UPLC (registered trademark) H-class biosystem (Waters Corporation: hereinafter also referred to as UPLC) was used for a liquid chromatography section of LC-MS/MS. As the column, an ACQUITY (registered trademark) UPLC (registered trademark) peptide BEH C.sub.18 column (Waters Corporation) which is a reversed-phase column was used. As the mass spectrometer, Xevo (registered trademark) TQ-XS triple quadrupole mass spectrometer (Waters Corporation: hereinafter also referred to as TQ-XS mass spectrometer) was used.

(70) Each eluate was placed on a UPLC autosampler, and 10 μ l of the eluate was introduced into the UPLC and fractionated by gradient. Conditions for the gradient were as follows.

(71) TABLE-US-00001 TABLE 1 Analysis device Xevo TQ-XS triple quadrupole mass spectrometer Column ACQUITY UPLC Peptide BEH C.sub.18 column (300 $\text{\AA}$, 1.7 μ m, 2.1 mm $\times$ 150 mm) Introduction amount 10 μ l Flow rate 200 μ l/min Temperature 50 $^{\circ}$ C. Mobile phase A 0.1% Ammonia solution Mobile phase B 0.01% Ammonia, 90% Acetonitrile solution Gradient conditions 0 to 1.0 minutes 90% A, 10% B 1.0 to 5.5 minutes 90-45% A, 10-55% B 5.5 to 6.7 minutes 45% A, 55% B 6.7 to 7.0 minutes 45-90% A, 55-10% B 7.0 to 8.5 minutes 90% A, 10% B

(72) An eluate that was subjected to the gradient and eluted from the column was directly subjected to the TQ-XS mass spectrometer connected to the UPLC. The TQ-XS mass spectrometer used electrospray ionization and measured in positive ion mode. Conditions for MRM measurement were set as shown in Table 2 below.

(73) TABLE-US-00002 TABLE 2 Cone Collision Precursor ion Product ion voltage energy (m/z) .Math. 4+ (m/z) .Math. 4+ (V) (eV) A β 40 1083.4 1054.0 32 22 A β 42 1129.5 1078.8 28 25 15N-A β 40 1096 1066.5 32 22 15N-A β 42 1142.6 1091.5 28 25

(3) Measurement Results

(74) Measurement results using LC-MS/MS are shown in FIGS. 2, 3A and 3B. FIG. 2 shows results of MRM measurement for A β 40 peptide, A β 42 peptide, 15N-A β 40 and 15N-A β 42, and FIGS. 3A and 3B show concentrations of A β 40 peptide and A β 42 peptide in the measured plasma specimens. From these results, it was shown that the A β peptide can be released from the complex using a basic solution containing an organic solvent and the A β peptide can be measured by mass spectrometry.

(75) FIGS. 4A and 4B show results of preparing a calibration curve based on the measured values detected using the A β peptide with a known concentration prepared in (1.3) above. As shown in FIGS. 4A and 4B, it was possible to prepare a calibration curve in which R_{sup.2} was 0.999 or more in both the A β 40 peptide and the A β 42 peptide. Low concentrations of A β peptide of 100 pg/ml or less were also detectable. From this, it was shown that the above measurement method which is a combination of a basic solution containing an organic solvent and mass spectrometry can measure A β peptide with high sensitivity and has excellent quantitative properties.

Example 2: Comparison of Elution Efficiency

(76) The composition of the free reagent was changed, and an elution efficiency of A β peptide from the complex due to difference in the composition of the free reagent was calculated based on the following calculation formula and compared.

[Elution efficiency (%)] = [A β peptide concentration of sample C] / ([A β peptide concentration of sample A] - [A β peptide concentration of sample B]) $\times$ 100

(1) Comparison of Basic Substances in Free Reagents

(1.1) Preparation of Free Reagent

(77) DDM (n-dodecyl- β -D-maltoside, Sigma-Aldrich Co. LLC.), acetonitrile and/or 28% concentrated ammonia water were appropriately selected and mixed with pure water so as to have the compositions shown in Table 3 below to prepare various free reagents. Here, a solution

containing no basic substance or organic solvent was also referred to as a free reagent for convenience. After preparation, pH of the solution containing the basic substance was measured using a pH meter (HORIBA, Ltd.).

(78) TABLE-US-00003 TABLE 3 Free reagent composition pH Comparative reagent 1 DDM, 70% Acetonitrile Comparative reagent 2 DDM, 0.056% Ammonia 11.013 Comparative reagent 3 DDM, 0.56% Ammonia 11.582 Reagent 1 50% Acetonitrile, 0.028% Ammonia 10.854 Reagent 2 50% Acetonitrile, 0.07% Ammonia 11.059 Reagent 3 50% Acetonitrile, 0.14% Ammonia 11.219 Reagent 4 50% Acetonitrile, 0.28% Ammonia 11.448 Reagent 5 50% Acetonitrile, 0.56% Ammonia 11.613 Reagent 6 50% Acetonitrile, 1.12% Ammonia 11.801 Reagent 7 50% Acetonitrile, 1.68% Ammonia 11.973 Reagent 8 50% Acetonitrile, 2.24% Ammonia 11.974 Reagent 9 50% Acetonitrile, 2.80% Ammonia 12.038

(1.2) Sample Preparation

(79) Sample A was prepared by suspending the A β 40 peptide used in (1.3) of Example 1 in a PBS solution containing 3% BSA so as to be 1000 pg/ml. The A β peptide concentration of sample A corresponds to an initial concentration of A β 40 peptide.

(80) (1.3) Immunoprecipitation

(81) The sample A prepared in (1.2) above was immunoprecipitated in the same manner as in (1.5) of Example 1 to recover magnetic particles. At this time, supernatant after magnetization was collected and stored as sample B. An A β peptide concentration of sample B corresponds to a concentration of A β peptide that could not be captured by an antibody that specifically binds to the A β peptide.

(82) (1.4) Washing/Elution

(83) The magnetic particles recovered in (1.3) above were washed in the same manner as in (1.6) of Example 1, and a washing liquid was removed. To the washed magnetic particles was added 15 μ l of each of the free reagents prepared in (1.1) above, and the mixture was allowed to stand for 1 minute to release the A β peptide. After standing the mixture, the magnetic particles were magnetically collected and supernatant was collected. A pH neutralizing solution (pH 7.4) containing 300 mM Tris and 300 mM NaCl was mixed with the collected supernatant to obtain sample C. The A β peptide concentration of the sample C corresponds to the concentration of A β peptide released after capture by immunoprecipitation.

(84) (2) Measurement by Immunoassay

(85) For the above samples A, B and C, the A β peptide concentration in each sample was measured by an immunoassay using a fully automated immunoassay system HISCL (registered trademark)-5000 (Sysmex Corporation). An R1 reagent (capture antibody reagent) was prepared by labeling 82E1 antibody with biotin by a conventional method and dissolving it in a buffer at pH 7.5 containing 1% BSA, 0.1 M Tris-HCl, 0.15 M NaCl and 0.1% NaN₃. As an R2 reagent (solid phase), a HISCL (registered trademark) R2 reagent (Sysmex Corporation) containing streptavidin-bound magnetic particles was used. An R3 reagent (detection antibody reagent) was prepared by labeling 1A10 antibody with alkaline phosphatase (ALP) by a conventional method and dissolving it in a buffer at pH 7.5 containing 1% BSA, 0.1 M Tris-HCl, 0.15 M NaCl and 0.1% NaN₃. As an R4 reagent (measurement buffer solution), a HISCL (registered trademark) R4 reagent (Sysmex Corporation) was used. As an R5 reagent (ALP substrate solution), a HISCL (registered trademark) R5 reagent (Sysmex Corporation) was used.

(86) Measurement procedure according to HISCL (registered trademark)-5000 was as follows. The sample A, B or C (30 μ L) and the R1 reagent (110 μ L) were mixed and reacted at 42° C. for 4 minutes. After the reaction, the R2 reagent (30 μ L) was added, and the mixture was reacted at 42° C. for 3 minutes. The magnetic particles in the obtained mixed solution were magnetically collected, supernatant was removed, and a HISCL (registered trademark) washing liquid (300 μ L) was added to wash the magnetic particles. Supernatant was removed, and the R3 reagent (100 μ L) was added to the magnetic particles and mixed, and the mixture was reacted at 42° C. for 5

50% acetonitrile (here, an acidic solution is also referred to as a free reagent for convenience). A free reagent (1.68% ammonia, 50% acetonitrile) was prepared in the same manner as in (1.4) of Example 1. To these free reagents was added the same A β 40 peptide as in (1.3) of Example 1 so as to have a final concentration of 100 fmol/ μ l (A β 40 peptide-containing solution). A solution containing no A β 40 peptide (A β 40 peptide-free solution) was also prepared for each of the acidic free reagent and the basic free reagent.

(96) (2) Introduction of Sample into Liquid Chromatography

(97) The four solutions prepared in (1) above were subjected to LC-MS/MS. As an LC-MS/MS apparatus and a column, those described in the mass spectrometry of (2) of Example 1 were used. Each solution was placed on the UPLC autosampler, and 10 μ l of each solution was introduced into the UPLC and fractionated. UPLC analysis conditions were as shown in Table 5 below. Conditions for MRM measurement in mass spectrometry are shown in Table 2 of Example 1.

(98) TABLE-US-00005 TABLE 5 Analysis device Xevo TQ-XS triple quadrupole mass spectrometer Column ACQUITY UPLC Peptide BEH C.sub.18 column (300 Å, 1.7 μ m, 2.1 mm $\times$ 150 mm) Introduction amount 10 μ l Flow rate 200 μ l/min Temperature 50° C. Mobile phase A 0.1% Ammonia solution Mobile phase B 0.01% Ammonia, 90% Acetonitrile solution Elution conditions 50% A, 50% B

(99) The two types of A β 40 peptide-containing solutions prepared in (1) above and the corresponding A β 40-free solutions thereof were subjected to UPLC, in the order of the A β 40 peptide-free solution (background), A β 40 peptide-containing solution, and A β 40 peptide-free solution (carryover measurement). The results of comparing signal intensities obtained at that time are shown in FIG. 7. From FIG. 7, it was found that a large amount of carryover occurred in a case where the acidic free reagent was used as compared with a case where the basic free reagent was used. On the other hand, when the basic free reagent was used as the free reagent for A β peptide, carryover could be remarkably suppressed. Therefore, it was shown that it is appropriate to use a basic free reagent for continuous analysis of A β peptide using mass spectrometry.

Example 4: Comparison of Free Reagents for Acidity and Basicity

(100) Carryover occurred when the A β peptide released using an acidic free reagent was measured by LC-MS/MS as in Example 3 above. Here, difference in the amount of A β peptide detected was compared between a case where a basic free reagent is used when releasing the A β peptide from the complex, and a case where an acidic free reagent is used when releasing the A β peptide, then replaced with a basic free reagent and subjected to LC-MS/MS.

(101) (1) Sample Preparation

(102) As the A β peptide sample, the same A β 40 peptide and A β 42 peptide as in (1.3) of Example 1 were used, and a solution prepared so that the A β 40 peptide was 50 pg/ml or 190 pg/ml in a 3% BSA solution, or a solution prepared so that the A β 42 peptide was 26 pg/ml or 103 pg/ml in a 3% BSA solution was prepared and used. As the acidic free reagent, trifluoroacetic acid (TFA, FUJIFILM Wako Pure Chemical Corporation) and acetonitrile were mixed with pure water to prepare an acidic solution having a composition of 0.1% TFA and 30% acetonitrile. As the basic free reagent, a 1.68% ammonia and 50% acetonitrile solution was prepared in the same manner as in (1.4) of Example 1.

(103) (2) Measurement

(104) (2.1) Immunoprecipitation

(105) An internal standard substance was added to each of the samples prepared in (1) above in the same manner as in (1.5) of Example 1, and magnetic particles immobilized with 6E10 antibody were added to form a complex.

(106) (2.2) Elution of A β Peptide

(107) Each complex prepared in (2.1) above was subjected to the same washing as in (1.6) of Example 1, and then 25 μ L of the acidic free reagent or basic free reagent prepared in (1.1) above was added and mixed, and the mixture was allowed to stand for 1 minute. The magnetic particles

were magnetically collected again, and supernatant was recovered as an eluate.

(108) (2.3) Solvent Exchange

(109) Of the eluates recovered in (2.2) above, the eluate obtained using the acidic free reagent was dried under reduced pressure for 1 hour under the conditions of a rotation speed of 1500 r/min and a temperature of 55° C. using Spin Dryer Standard VC-96R (TAITEC CORPORATION) in SpinDryer mode to volatilize the solvent. To a residue after drying under reduced pressure, 25 μ L of the basic free reagent prepared in (1) above was added to resuspend the residue.

(110) (2.4) Mass Spectrometry

(111) The eluate (referred to as solution X) obtained by using each basic free reagent recovered in (2.2) above and the solution (referred to as solution Y) resuspended in (2.3) above were subjected to LC-MS/MS in the same manner as in (2) of Example 1, and MRM measurement was performed.

(112) (3) Measurement Results

(113) Measurement results of solution A and solution B are shown in FIGS. 8A to 9D, respectively. FIG. 8A shows a measurement result of solution X using 190 pg/ml A β 40 peptide, FIG. 8B shows a measurement result of solution Y using 190 pg/ml A β 40 peptide, FIG. 8C shows a measurement result of solution X using 103 pg/ml A β 42 peptide, FIG. 8D shows a measurement result of solution Y using 103 pg/ml A β 42 peptide, FIG. 9A shows a measurement result of solution X using 50 pg/ml A β 40 peptide, FIG. 9B shows a measurement result of solution Y using 50 pg/ml A β 40 peptide, FIG. 9C shows a measurement result of solution X 26 pg/ml A β 42 peptide, and FIG. 9D show a measurement result of solution Y using 26 pg/ml A β 42 peptide. From these results, in the examples of A β peptides at all concentrations, the area value was larger when using the solution X than when using the solution Y. It was suggested that loss of A β peptide occurred when the process as described in (2.3) above was used in sample treatment.

Claims

1. A method for measuring an A β peptide, comprising: mixing i) a blood sample comprising an A β peptide and ii) an antibody that specifically binds to the A β peptide, to form a complex of the A β peptide and the antibody; releasing the A β peptide from the complex with a basic solution comprising acetonitrile and ammonia; and measuring a released A β peptide in the basic solution by liquid chromatography-mass spectrometry, wherein the basic solution containing the released A β peptide is used as is for liquid chromatography-mass spectrometry, and wherein the basic solution has a pH of 11.4 or more.
2. The method according to claim 1, wherein the basic solution further comprises at least one selected from the group consisting of acetone, 1-propanol, 2-propanol, hexane, ethanol, dimethyl sulfoxide and methanol.
3. The method according to claim 1, wherein the antibody that specifically binds to the A β peptide is an antibody that binds to one or both of A β 40 and A β 42, and the A β peptide released from the complex in the releasing is at least one of A β 40 and A β 42, and a measured value of the at least one of A β 40 and A β 42 is acquired in the measuring.
4. The method according to claim 3, wherein the antibody that specifically binds to the A β peptide is an antibody that binds to both A β 40 and A β 42, and respective measured values of A β 40 and A β 42 are acquired in the measuring.
5. The method according to claim 1, wherein the antibody that specifically binds to the A β peptide is immobilized on a solid phase.
6. The method according to claim 1, wherein a reversed-phase column is used in the liquid chromatography-mass spectrometry.
7. The method according to claim 1, wherein the blood sample is mixed with a labeled internal standard substance, and in the measuring, a measured value of the AB peptide is acquired based on a measured value of the internal standard substance.

8. The method according to claim 7, wherein the internal standard substance is an A β peptide labeled with a stable isotope.

9. A method for measuring an A β peptide, comprising: mixing i) a blood sample comprising an A β peptide and ii) an antibody that specifically binds to the A β peptide, to form a complex of the A β peptide and the antibody; conducting bound/free separation to remove an unreacted free component that has not formed the complex and recover the complex; releasing the A β peptide from the complex with a basic solution comprising acetonitrile and ammonia; and measuring a released A β peptide in the basic solution by liquid chromatography-mass spectrometry, wherein the basic solution containing the released A β peptide is used as is for liquid chromatography-mass spectrometry, and wherein the basic solution has a pH of 11.4 or more.

10. The method according to claim 9, wherein the basic solution further comprises at least one selected from the group consisting of acetonitrile, acetone, 1-propanol, 2-propanol, hexane, ethanol, dimethyl sulfoxide and methanol.

11. A method for measuring an A β peptide, comprising: mixing i) a blood sample comprising an A β peptide, ii) an antibody that specifically binds to the A β peptide and iii) a labeled internal standard substance, to form a complex of the A β peptide and the antibody and a complex of the labeled internal standard substance and the antibody, wherein the labeled internal standard substance is a labeled A β peptide; conducting bound/free separation to remove an unreacted free component that has not formed the complex of the A β peptide and the antibody or the complex of the labeled internal standard substance and the antibody; releasing with a basic solution comprising acetonitrile and ammonia the A β peptide from the complex of the A β peptide and the antibody, and releasing with the basic solution the labeled internal standard substance from the complex of the labeled internal standard substance and the antibody; and measuring a released A β peptide and the labeled internal standard substance by liquid chromatography-mass spectrometry and quantifying the A β peptide based on a measured value of the labeled internal standard substance, wherein the basic solution containing the released A β peptide is used as is for liquid chromatography-mass spectrometry, and wherein the basic solution has a pH of 11.4 or more.

12. The method according to claim 11, wherein the basic solution further comprises at least one selected from the group consisting of acetonitrile, acetone, 1-propanol, 2-propanol, hexane, ethanol, dimethyl sulfoxide and methanol.

13. The method according to claim 11, wherein the internal standard substance is an A β peptide labeled with a stable isotope.
